



ENERGY MIX SIMULATOR

Exploring the paths of energy transition

theme: energy and energy efficiency



Introduction

Our energy system relies on a complex balance between different energy sources. In 2023, the EU produced 2,572 TWh (terawatt hours) of electricity. More than 45% of this production came from renewable sources. Fossil fuels accounted for 31.7% and nuclear electricity nearly 23%. Gas was the main fossil fuel used to generate electricity (17%), followed by coal (11.7%).

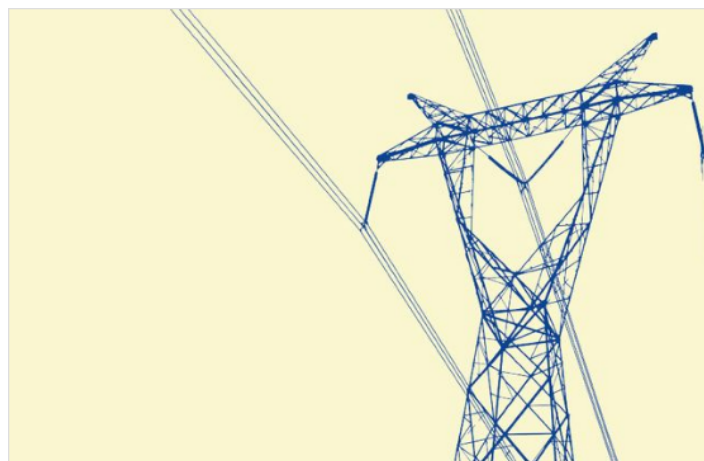
The electricity mix varies considerably between Member States, with the share of electricity generated from renewable sources ranging from over 90% to less than 15%.

This diversity is not accidental: it results from strategic choices aimed at reconciling security of supply, acceptable costs and climate objectives.

However, this interdependence also creates vulnerabilities. The 2022-2023 energy crisis, triggered by the war in Ukraine, revealed the consequences of excessive dependence on Russian gas. Similarly, the planned shutdowns of nuclear power plants in Germany and Belgium raise questions about the system's ability to maintain a balance between supply and demand while meeting climate targets.

The International Energy Agency (2023) emphasizes that the energy transition requires rigorous planning: each change in the energy mix leads to cascading effects on costs, emissions, security of supply, and social acceptability. These complex interactions cannot be understood intuitively and require modeling tools.

This protocol invites students to build their own energy mix simulator to explore the consequences of completely or partially eliminating an energy source. Through this modeling approach, they will develop a systemic understanding of energy issues and the constraints facing political and economic decision-makers. The goal is not to train technical experts, but to develop their ability to analyze complex systems and anticipate the consequences of seemingly simple decisions.



Disciplines

Sustainable Development Goals



physics
history-geography
mathematics





Overview

Protocol structure

This protocol aims to develop in students a deep understanding of the issues of energy interdependencies through the construction and use of a simulation tool. The pedagogical approach is based on quantitative modeling, numerical experimentation, and the analysis of prospective scenarios.

The activity is structured around four progressive phases:

1. **Step 1: Deciphering the energy mix** - Students analyze the current composition of their country's electricity mix, identify the characteristics of each energy source (cost, controllability, emissions, constraints) and understand the specific functions of each technology in the system.
2. **Step 2: Building the Simulator** - In groups, students build a simulation tool to model the economic, environmental, and technical impacts of changes to the energy mix. The simulator includes calculation formulas, dynamic graphs, and performance indicators.
3. **Step 3: Experimentation and Crisis Scenarios** - Students test their simulator by exploring the consequences of phasing out or completely phasing out different energy sources (gas, nuclear, coal, intermittent renewables). They analyze the impacts on costs, emissions, and security of supply. Based on their simulations, students formulate recommendations for managing energy transitions, identifying optimal sequences for transforming the mix and the necessary support measures.

This progression allows students to move from a static understanding of the energy mix to a dynamic vision of transition issues, by developing their capacity for modeling and prospective analysis.

Getting started

Duration: min. 4 sessions

Level of difficulty: Advanced - Requires skills in mathematical modeling, data analysis, and spreadsheet programming to build the simulator)



Material needed:

- Step 1: Internet access for data research, documents on the European energy mix, calculators
- Step 2: Computers with Excel or equivalent (LibreOffice Calc), pre-formatted simulator models, Excel construction guide
- Step 3: Printed scenario sheets, access to the created simulators, material for recording results, presentation materials (video projector, tables), material for creating visual recommendations

Glossary

Keywords and concepts	Definitions
Energy mix	Distribution of different energy sources in the electricity production of a territory, expressed as percentages.
Base load (charge de base)	Minimum level of electrical demand on a network during a given period, generally provided by controllable sources.
Controllable sources	Production technologies whose power can be adjusted according to demand (nuclear, gas, coal, hydraulic).
Intermittent sources	Production technologies dependent on variable natural conditions (wind, solar photovoltaic).
Load factor	Ratio between the actual production of an installation and its maximum theoretical production over a given period.
LCOE (Levelized Cost of Energy)	Average cost of producing a unit of electricity over the lifetime of an installation, including investment and operation.
Installed capacity	Theoretical maximum electrical power of an installation or production park, expressed in MW or GW.
Electrical network	Electricity transmission and distribution infrastructure, which must maintain the supply-demand balance in real time.
Energy storage	Technologies that allow energy to be stored for later use (batteries, STEP, hydrogen).
Interconnections	Power lines connecting the networks of different countries, allowing the exchange of electricity.
Avoided emissions	Quantity of CO ₂ not emitted by replacing a fossil source with a low-carbon source.
Security of supply	The ability of an energy system to meet demand continuously, even in the event of equipment failure.
Merit order	Order of priority for calling power plants according to their marginal production cost, from least expensive to most expensive.
Energy mix	Distribution of different energy sources in the electricity production of a territory, expressed as percentages.
Base load (charge de base)	Minimum level of electrical demand on a network during a given period, generally provided by controllable sources.



Protocol

Step 1 - Deciphering the energy mix



Background and description of the problem to be solved at this stage: Before being able to model the impacts of changes in the energy mix, students must understand the current logic of their country's electricity system. This phase leads them to decipher the technical, economic, and environmental characteristics of each energy source to understand why the current mix has formed as it has and what constraints are weighing on its evolution.

Learning Objectives: Analyze the composition of the current energy mix. Identify the specific characteristics of each energy source. Understand the complementary functions of different technologies in the electrical system.

Conceptualisation



Research question: What factors determine the current composition of the European energy mix and national specificities?

- **Hypothesis:** The composition of the energy mix results from a complex balance between geographical constraints (available resources), historical legacies (existing infrastructures), political choices (energy strategies) and economic constraints (production costs), creating national specificities within common European objectives.
- **Key concepts:**
 - Optimal energy mix: Distribution of sources aimed at simultaneously optimizing cost, safety and sustainability (reveals the necessary trade-offs)
 - Path dependence: Influence of past choices on current options for system evolution (explains system inertia)
 - Systemic constraints: Technical and economic limitations which condition energy choices (determine room for maneuver)



Research question: How do the specific technical characteristics of each energy source determine their complementary role in the electricity system?

- **Hypothesis:** Each energy source has unique technical characteristics (controllability, flexibility, load factor) which make it suitable for specific functions in the electrical system, making simple and direct substitution between technologies impossible.
- **Key concepts:**
 - Functional complementarity: Need to combine different sources to ensure all the functions of the electrical system (base, semi-base, peak, adjustment)
 - Controllability vs. intermittency: Fundamental opposition between controllable sources and sources dependent on natural conditions (structures network management issues)
 - Order of merit: Order of calling of power plants according to their marginal cost which determines their effective use (explains the economic hierarchy of sources)



Research question: What tensions exist between economic, environmental and security of supply objectives in European energy choices?

- **Hypothesis:** European energy choices result from complex trade-offs between three sometimes contradictory objectives: minimizing costs (economic competitiveness), reducing environmental impacts (climate objectives)

and guaranteeing security of supply (geopolitical independence).

- Key concepts:
 - Energy trilemma: Tension between affordability, sustainability and energy security that constrains political choices (structures energy debates)
 - Environmental externalities: Environmental costs not integrated into market prices (explains distortions between private and social costs)
 - Energy sovereignty: Ability to maintain supply regardless of geopolitical crises (influences national and European strategies)

Students Investigation

1. Overview of the national energy mix



Objective: To analyze the current composition of the national energy mix and identify development trends.

Students receive recent data on their country's electricity production and work in pairs to analyze this information. They complete a summary table that will form the basis of their future simulator. Students calculate totals, check data consistency, and identify trends. They then create a pie chart showing the current distribution of the mix.

Analysis questions:

- Which sources dominate the current mix and why?
- Which sources are growing fastest and what factors explain this development?
- What is the share of controllable vs. intermittent sources?

2. Analysis of technical and economic characteristics



Objective: To understand the specificities of each energy source which determine its role in the electrical system.

Each pair specializes in two or three energy sources and completes a detailed analysis sheet. These sheets will then serve as the database for building the simulator.

Analysis sheet template (example: Natural gas) - Available for printing in the appendix

TECHNICAL DATA SHEET - NATURAL GAS	
Technical characteristics	
Average load factor	45%
Start-up time	15-30 minutes
Flexibility	Very high (0-100% in a few minutes)
Installation lifespan	25-30 years old
Economic characteristics	
Investment cost	800-1200 €/kW
Operating cost	60-80 €/MWh
Average LCOE	75-95 €/MWh
Environmental characteristics	
CO2 emissions	350-400 g/kWh
Other impacts	NOx, dependence on imports
Benefits in the system	Disadvantages in the system
Flexibility to balance the network	Volatile fuel prices
Fast response time	Geopolitical dependence
Mastered technology	Significant CO2 emissions

Technical characteristics:

- The load factor indicates the ratio between actual production and theoretical maximum production. It is crucial for calculating effective production.
- Start-up time and flexibility determine the ability to respond to variations in demand
- Lifespan influences investment depreciation

Economic characteristics:



- The investment cost (€/kW) allows the total installation cost to be calculated
- The operating cost (€/MWh) determines the variable production costs
- LCOE combines all costs to give the average cost of production over the lifetime

Environmental characteristics:

- CO2 emissions (g/kWh) are used to calculate the total climate impact
- Other environmental impacts must also be considered in the overall assessment

These parameters are essential for modeling the technical, economic and environmental impacts of any change in the energy mix.

Students complete similar sheets for all sources in the mix, drawing on provided documentary resources and guided research.

3. Functions in the electrical system

Students collectively analyze how different energy sources play complementary and essential roles in their national energy system. This understanding is crucial because each type of power plant meets specific grid needs:

Functional distribution of the mix:

- **Base load (40-60% base load):** Nuclear, run-of-river hydro, share of coal - These sources operate continuously 24 hours a day to provide the basis for electricity production, with high fixed costs but low variable costs
- **Semi-base (average load 20-30%):** Base gas, lake hydro - These plants adapt their production according to daily variations in demand, with moderate flexibility and average variable costs
- **Peak (peak load 10-20%):** Flexible gas, pumped hydraulics, imports/exports - These highly responsive units can start up quickly to meet peak consumption, despite high production costs
- **Weather-dependent:** Wind and solar, offset by other sources - These renewable energies produce according to weather conditions, requiring other sources to adapt to maintain grid balance

This analysis is fundamental for the construction of the simulator because it highlights that each source has unique technical and economic characteristics: start-up time, flexibility, costs, etc. These constraints limit the possibilities of substitution between sources, because they cannot all ensure the same functions in the electrical system.

Conclusion & Further Reflexion

Analyzing the European energy mix revealed to students the complexity of the trade-offs underlying current energy choices. They learned that each energy source has specific technical, economic, and environmental characteristics that determine its role in the electricity system. This functional complementarity explains why no single source can meet all the network's needs and why energy transitions require complex restructuring rather than simple substitutions.

The detailed datasheet approach encourages the gradual appropriation of energy concepts and develops rigor in data analysis. Specialization by group and then sharing allows students to delve deeper into certain aspects while building an overall vision. The functional analysis of the electrical system helps students move beyond a static vision to understand dynamic interdependencies.

Students now have a grasp of the key characteristics of different energy sources and understand how they fit into the European electricity system. They have developed their energy data analysis skills and acquired the conceptual foundations needed to model the impacts of changes in the energy mix.

To further the reflection and prepare for the construction of the simulator, here are some key questions to explore collectively:

- **What are the physical and economic limits of energy source substitutions?** Identifying the technical constraints that prevent certain substitutions (intermittency, response time, location). Analyzing the hidden costs of transitions (infrastructure, storage, flexibility). Understanding the critical thresholds for the stability of the electricity system.
- **How can we anticipate the future evolution of the energy mix?** Exploring current trends (growth of renewables, decline of fossil fuels, debates on nuclear power). Identifying factors accelerating or slowing transitions. Reflecting on technological innovations likely to transform the energy landscape.
- **What modeling tools should be developed to assess the impacts of changes in the mix?** Preparation for building the simulator by identifying the key variables to integrate. Consideration of relevant indicators for evaluating scenarios (cost, emissions, safety). Introduction to simulation and optimization methods applied to energy systems.
- **How do national specificities fit with common European objectives?** Analysis of the differences in energy mix between European countries and their causes (geography, history, political choices). Exploration of European coordination mechanisms and energy solidarity issues. Reflection on the balance between national sovereignty and European integration.

Step 2 - Construction of the simulator



Background and description of the problem to be solved at this stage: Armed with their understanding of the current energy mix, students will now build a tool allowing them to model the impacts of changes to this mix. This sequence develops their numerical modeling skills while deepening their understanding of energy interdependencies.

Learning objectives: Build a digital model of the European energy mix. Integrate technical and economic constraints into the modeling. Develop performance indicators to evaluate different scenarios.

Conceptualisation



Research question: How to translate the complexity of the energy system into an accessible and functional digital model?

- Hypothesis: Modeling an energy system requires simplifying reality by preserving the essential relationships between technical, economic and environmental variables, while making the tool usable by non-experts to explore different scenarios.
- Key concepts:
 - Systemic modeling: Simplified representation of a complex system while retaining its essential properties (allows analysis and prediction)
 - Endogenous vs. exogenous variables: Distinction between user-modifiable parameters and imposed external constraints (structures the model architecture)
 - Model validation: Verification of consistency between simulated results and real data (guarantees the reliability of analyses)



Research question: What indicators and constraints should be integrated to appropriately assess energy scenarios?

- Hypothesis: The evaluation of energy scenarios requires multidimensional indicators (economic, environmental, technical) and realistic constraints that reflect the physical and political limits of the system, to avoid unfeasible theoretical solutions.
- Key concepts:
 - Energy performance indicators: Quantitative metrics for assessing the quality of an energy mix according to different criteria (guides strategic choices)
 - Feasibility constraints: Technical, economic or political limits that limit the space of possible solutions (avoids unrealistic scenarios)
 - Multi-criteria optimization: Search for solutions balancing several sometimes contradictory objectives (reflects the complexity of energy choices)



Research question: How does the collaborative construction of a simulation tool promote the appropriation of energy issues?

- Hypothesis: Collaborative construction of a simulator develops a deep understanding of energy mechanisms because it requires explaining causal relationships, quantifying impacts and comparing hypotheses with real data.
- Key concepts:
 - Learning by modeling: Process of acquiring knowledge by constructing a formalized representation (promotes systemic understanding)

- Technical appropriation: Mastery of a tool through its construction rather than its passive use (develops analytical autonomy)
- Computational thinking: Ability to decompose a complex problem into modelable and calculable elements (transferable skill)

Students Investigation

1. Simulator architecture



Objective: Design the general structure of the simulator and organize the basic data.

Students, in groups, receive a file (e.g., Excel or Google Sheet) pre-formatted with basic tabs and gradually build their simulator. The teacher guides the construction step by step.

Example of file structure

Tab 1: "Basic Data"

- Table of characteristics of each source (data collected in Sequence 1)
- Reference parameters (total demand, climate objectives, budgets)
- Conversion factors and constants

Tab 2: "Current Mix"

- Current distribution of the mix (in % and in TWh)
- Automatic calculations of costs, emissions and technical indicators
- Visualization charts

Tab 3: "Simulator"

- Mix Edit Interface
- Real-time calculations of impacts
- Alerts on constraints not respected

Tab 4: "Results"

- Before/After Comparison Chart
- Impact graphs
- Summary indicators

Students begin by entering their basic data in Tab 1, organizing the information collected during the previous sequence.

2. Construction of calculation formulas



Objective: Develop calculation algorithms to quantify the impacts of mix changes.

The teacher guides the students in the progressive construction of the simulator's key formulas:

Production by source



$$\text{Production_source} = \text{Part_mix} \times \text{Total_demand} \times \text{Load_factor}$$

This formula calculates the actual production of each energy source by taking into account its share in the mix, the total electricity demand, and the load factor which represents the average availability of the source (for example, solar does not produce at night).

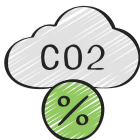
Total system cost



$$\text{Total_cost} = \Sigma(\text{Installed_capacity} \times \text{Investment_cost} + \text{Production} \times \text{Operating_cost})$$

This formula adds up the fixed costs (initial investment multiplied by installed capacity) and variable costs (operating costs multiplied by production) for each source. This gives the total annual cost of the electricity system.

Total emissions



$$\text{Total_emissions} = \Sigma(\text{Production_source} \times \text{Source_emission_factor})$$

This formula calculates total CO2 emissions by multiplying each source's output by its specific emission factor (in gCO2/kWh). Some sources, such as nuclear or hydropower, have very low factors, while coal has a high factor.

Security indicator



$$\text{Controllable_part} = \Sigma(\text{Controllable_sources_production}) / \text{Total_production}$$

This formula calculates the proportion of electricity from controllable sources (nuclear, thermal, dam hydro) compared to total production. A high ratio indicates better grid stability.

Autonomy indicator



$$\text{Import_rate} = \text{Import_production} / \text{Total_demand}$$

This formula assesses import dependence by calculating their share in meeting total demand. A lower rate indicates greater energy autonomy.

Students implement these formulas step by step, test their operation and add checks (sum of parts = 100%, consistency of units).

3. User interface and visualization



Objective: Create an intuitive interface allowing you to easily modify the mix and visualize the results.

Students develop their simulator interface to make it easy to use:

- **Identification of input cells to modify the share of each source**
- **Automatic constraint: the sum must equal 100%**
- **Colored alerts: red if constraints violated, green if objectives achieved**
- **Dynamic graphics: mix pie chart, cost/emissions bars**
- **Dashboard: key indicators highlighted**

The groups test their simulator by changing a few parameters and verifying that the calculations and graphs update correctly.



Example dashboard

KEY INDICATORS	Current Mix	New mix	Evolution
Total cost (€ billion/year)	180	=FORMULA	=FORMULA
Emissions (Mt CO2/year)	950	=FORMULA	=FORMULA
Controllable share (%)	61%	=FORMULA	=FORMULA
Renewable share (%)	39%	=FORMULA	=FORMULA

4. Validation and calibration

Students calibrate their simulator by comparing its results with reference data and adjusting the parameters if necessary. They also prepare simple user instructions for the following sequences.

Conclusion & Further Reflexion

Building the simulator allowed students to translate their conceptual understanding of the energy mix into a functional digital model. They learned to structure complex data, formalize causal relationships with mathematical formulas, and create interfaces for interactive scenario exploration. This approach introduced them to the challenges of modeling: balancing simplicity and realism, validating results, and managing multiple constraints.

The step-by-step approach promotes technical and conceptual understanding of energy issues. Working in small groups allows for mutual assistance and the comparison of approaches. Validation through comparison with real data develops critical thinking and scientific rigor. The final user interface gives students the satisfaction of creating a functional and usable tool.

Students now have a solid grasp of the quantitative mechanisms that govern energy systems (calculating production, costs, emissions, and performance indicators). They have developed their skills in numerical modeling and their ability to translate complex concepts into operational tools. They have an analytical tool they can use to explore different energy transition scenarios.

To deepen the reflection and prepare for the experimentation of scenarios, here are some key questions to explore collectively:

- **What are the limitations of our model and how can we take them into account when interpreting the results?** Identify the simplifications made and their impact on the validity of the analyses. Reflect on the gap between the model and reality. Discuss the necessary precautions when using simulation results.
- **How can our simulator shed light on current energy debates?** Exploration of contemporary energy issues that the simulator can analyze. Identification of stakeholders (politicians, businesses, citizens) who could benefit from such tools. Reflection on the role of modeling in decision-making.
- **Which crisis or transition scenarios would be the most instructive to test?** Anticipate the experiments in the next sequence by identifying the situations that most reveal energy issues. Reflect on geopolitical, climatic, or technological events likely to disrupt the energy mix.
- **How can we gradually improve our simulator to make it more realistic?** Identifying possible refinements (taking into account seasonal variations, network costs, technological innovations). Reflecting on the balance between increasing complexity and ease of use. Preparing for the tool's continued evolution.

This phase provides students with the technical tools necessary to become independent analysts of energy issues, capable of questioning political and media discourses using rigorous quantitative analyses.

Step 3 - Experimentation and energy transition scenarios



Background and description of the problem to be solved at this stage: Students now have a functional tool to explore the consequences of changes in the energy mix. This phase leads them to test different scenarios involving the elimination or drastic reduction of an energy source, then to formulate realistic strategic recommendations. The objective is to understand the complexity of energy transitions and to develop their synthesis skills to propose operational solutions.

Learning objectives: Analyze the impacts of eliminating a major energy source. Identify compensation strategies and their limitations. Assess the technical and economic feasibility of different transitions. Formulate strategic recommendations for European decision-makers.

Conceptualisation



Research question: What are the physical constraints and limits that condition the feasibility of major energy transitions?

- **Hypothesis:** Energy transitions are limited by physical (deployment time, industrial capacities, network stability) and economic (investment costs, social acceptability) constraints which make certain rapid substitutions between energy sources impossible, requiring progressive and diversified transition strategies.
- **Key concepts:**
 - Systemic inertia: Resistance of the energy system to rapid changes due to existing infrastructure and technical constraints (explains incompressible delays)
 - Breaking points: Thresholds beyond which technical or economic constraints make a transition impossible (delimits the space of feasible solutions)
 - Transition path: Optimal sequence of energy mix changes respecting all constraints (guides strategic planning)



Research question: How do compensation strategies reveal interdependencies and complementarities between energy sources?

- **Hypothesis:** The elimination of a major energy source cannot be compensated by a single alternative but requires a complex recomposition of the mix, revealing that each source has specific irreplaceable functions and that energy security relies on diversification.
- **Key concepts:**
 - Imperfect substitution: Impossibility of directly replacing one energy source with another without modifying the other components of the system (reveals the complexity of transitions)
 - System effects: Cascading impacts of a change on the entire energy mix (demonstrates interdependencies)
 - Energy portfolio: Risk diversification approach applied to the energy mix to ensure security of supply (justifies the need for multiple sources)



Research question: What trade-offs between conflicting objectives structure energy policy choices?

- **Hypothesis:** Energy decisions result from complex trade-offs between often contradictory objectives (cost, climate, security, social acceptability) which explain why there is no single optimal solution but only compromises reflecting political and social priorities.
- **Key concepts:**
 - Energy trilemma: Permanent tension between affordability, sustainability and security which constrains all energy choices (structures political debates)

- Optimization under multiple constraints: Search for solutions that simultaneously satisfy several sometimes incompatible objectives (reflects the complexity of public decisions)
- Energy governance: Collective decision-making process integrating technical expertise and democratic legitimacy (questions the modes of citizen participation)

Students Investigation

1. Phase-out scenarios and compensation strategies



Objective: To methodically explore the consequences of gradually reducing an energy source and test different strategies to compensate for the removed energy source while respecting realistic constraints.

The students collectively analyze the "transport budget" of a typical European family, then explore different levers of sobriety.



Group 1 - Coal Exit

"Following the Paris Agreement and public pressure, your country is committed to phasing out coal within five years to meet its climate targets."

- Reduce the share of coal from 16% to 0%
- 5-year transition (urgent climate objective)
- Offset 400 TWh/year of controllable production

”



Group 2 - Exit from nuclear power

"Following a national referendum, your country has decided to phase out nuclear power over the next 15 years, closing the power plants at the end of their lifespan."

- Reduce the share of nuclear power from 23% to 0%
- Transition in 15 years (lifespan of power plants)
- Offset 580 TWh/year of low-carbon and controllable production

”



Group 3 - Gas Crisis

"A major geopolitical crisis is cutting off your country's natural gas supply. You must drastically reduce this dependence within two years."

- Reduce the share of gas from 22% to 5%
- Transition in 2 years (supply crisis)
- Offset 425 TWh/year of flexible production

”



Group 4 - Limitation of renewables

"Grid managers are warning of growing instability due to intermittent renewables. Their development must be capped."

- Cap wind and solar at 25%
- Limited local acceptability
- Meeting climate targets without intermittent overcapacity

”

They use their simulator to analyze the impacts of different transition rates. The goal is not to take a position for or against these scenarios, but to understand their technical and economic implications. The groups must use the simulator correctly, interpret the results regarding costs, emissions, and safety, note the simulator's warning messages, and document their observations.

The groups then identify the optimal pace and prepare a summary of their findings, before presenting the main lessons in two minutes.

For each scenario, students systematically explore the different compensation options and identify the breaking points:

Strategy	Benefits	Limits	Relative cost
Natural gas increase	Flexibility, rapid deployment	CO2 emissions, import dependence	Average
Wind development	Low carbon, decreasing cost	Intermittency, acceptability	Weak
Solar development	Low-carbon, modular	Intermittence, storage required	Weak
Nuclear expansion	Low-carbon, controllable	Long lead times, high costs	Pupil
Imports	Speed, flexibility	Dependence, transport costs	Variable
Energy sobriety	Reduction in aggregate demand	Acceptability, physical limits	Very weak

Constraints to be integrated



- Controllability limit: Minimum 50% of controllable sources for network security Simple explanation: "If too much electricity comes from wind and sun, the network becomes unstable"
- Budget limit: Maximum +50% cost compared to the current mix Simple explanation: "Citizens and industries cannot pay twice as much for electricity"
- Climate limit: Maximum 150g CO2/kWh (European 2030 target) Simple explanation: "We must respect Europe's climate commitments"
- Import limit: Maximum 15% of demand (energy security) Simple explanation: "Being too dependent on neighbors is risky in the event of a crisis"
- Deployment limit: Maximum +10% per year for renewables (industrial capacities) Simple explanation: "We cannot build wind turbines faster than industry can produce them"

The groups freely explore different combinations in their simulator to compensate for the removed energy source. The groups test the limits of their strategies.

The teacher guides: "What happens if you want to go faster? Further?" It is necessary to identify precisely where and why the system "breaks." For example, Group 2 (nuclear phase-out) may discover that replacing 100% of nuclear power with wind power would require increasing current capacity eightfold, which would take 22 years at the maximum rate of +10%/year, making their 15-year goal impossible without other sources.

Finally, each group selects its optimal strategy, justifies its choices and prepares an argument for the next step.

2. Formulation of strategic recommendations



Objective: To translate lessons learned from simulations into strategic principles and operational roadmaps.

This final activity transforms students from simulator users into strategic advisors. The goal is for them to synthesize their findings to formulate realistic recommendations.

Comparative analysis of scenarios: The groups present their results in a standardized format that the teacher can project in the classroom if possible.

Evaluation criteria	Coal outlet	Nuclear exit	Gas crisis	Renewal limitation.
Technical feasibility	✓ Possible	⚠ Difficult	✓ Possible	✓ Possible
Cost impact	+12%	+25%	+18%	+8%
Impact emissions	-45%	+15%	-20%	+30%
Security of supply	✓ Maintained	⚠ Weakened	⚠ Weakened	✓ Reinforced
Deadline for completion	5 years	15 years old	2 years	Immediate
Social acceptability	✓ Strong	⚠ Debated	✓ Necessary	⚠ Conflictual

Technical feasibility is assessed positively if no major constraints are violated, with an alert otherwise. The cost impact corresponds to the percentage increase compared to the current mix. The emissions impact represents the percentage reduction or increase in CO2 emissions. Safety is assessed positively if more than 50% of controllable sources and less than 15% of imports are maintained. The acceptability criterion assesses whether the population broadly supports the transition (popular) or whether it arouses opposition (controversial).

Formulation of strategic principles: Based on their simulations, students collectively formulate strategic principles:

- Principle of progressiveness: Abrupt transitions create imbalances that are difficult to manage
- Principle of advance compensation: Alternatives must be developed before existing sources are eliminated
- Principle of diversification: No single source can replace another in an equivalent manner
- Principle of acceptability: Technical transitions must be accompanied by social consensus
- Resilience principle: The mix must maintain its capacity to adapt to crises

Roadmap Design: Each group designs a synthetic “roadmap” for its transition, applying the identified principles:

- Objective: Targeted reduction and timeframe
- Main phases: 3 steps with key actions
- Budgets and profits: Orders of magnitude
- Critical risks: Main points of vigilance

3. Debate and political arbitrations



Objective: Simulate collective decision-making processes and understand the complexity of energy choices.

Based on the roadmaps, the teacher proposes a simulation of an "Energy Council" where each group will have to defend its proposal before the Council.

The students will play a dual role, both as experts during the presentation of their roadmap, then as “advisors” according to a role which will be allocated to them by drawing lots:

- Citizens' Representative: Priority to energy costs and local employment
- Representative of industrialists: Priority to competitiveness and security of supply
- Green Representative: Priority to the climate and the phasing out of fossil fuels
- Union representative: Priority to employment and training of workers



Note for spatial organization: The room should be organized from the beginning of the “Debate” activity into four zones corresponding to the sub-councils (citizens, industrialists, environmentalists, unions). Students work within their sub-councils throughout the session, reforming their expert groups only briefly during the presentations on the board that begin the session. This spatial organization reinforces immersion in the roles of representatives and promotes cohesion within each sub-council throughout the debate.

The first phase of the "Debate" activity consists of presenting the four roadmaps. Each group takes turns presenting their conclusions for five minutes, clearly explaining the objectives, implementation steps, costs, and expected benefits. During these presentations, the other students prepare their presentations based on their specific roles.

The debate phase is organized around four strategic questions that the teacher asks successively.

- **"Which energy transition should be considered most urgent and why?"** Each group of advisors expresses its position according to its specific concerns, naturally creating tensions between the climate emergency, economic constraints, and social issues.
- **"How can we concretely finance the investments needed for these transitions?"** The teacher guides the discussion through the different options (taxes, public borrowing, European funding, private participation), asking each group to respond to the acceptability of these mechanisms for their representatives.
- **"What strategy should be adopted to manage the social consequences of these transitions?"** This question allows us to address the issues of professional retraining, supporting changing territories, and social justice in the energy transition.
- **"How much uncertainty and risk are we willing to accept regarding our energy security?"** This discussion reveals the trade-offs between prudence and ambition, between national autonomy and European cooperation.

The teacher plays an active role as a moderator by reformulating positions to avoid misunderstandings, relaunching the debate when it runs out of steam, and pushing for concrete compromises. He or she can directly challenge certain groups: "Would the unions accept this proposal if it were accompanied by a training plan?" or "Are manufacturers prepared to pay a carbon tax if it finances infrastructure?"

The conclusion of the debate aims to bring out the collective lessons learned from this negotiation experience. The instructor asks each group of advisors to identify a point of convergence discovered during the discussions and a concrete proposal to overcome the identified obstacles. This summary generally reveals that initial positions can evolve when the participants understand each other's constraints and seek solutions together.

Conclusion & Further Reflexion

Experimenting with transition scenarios revealed to students the complexity of energy choices and the existence of physical and economic constraints that limit the available options. They discovered that energy transitions cannot be thought of as simple substitutions but require systemic recompositions respecting multiple constraints. The formulation of strategic recommendations allowed them to understand that energy choices result from trade-offs between often contradictory objectives and require political and social compromises.

The interactive simulation approach encourages students to take ownership of the issues and assess the impact of their choices in real time. Facing realistic constraints develops their understanding of the limits of what is possible and their ability to adapt. The final role-playing debate anchors learning in a civic perspective and develops their argumentation and negotiation skills.

Students now have a grasp of the mechanisms that govern energy transitions and are able to assess the feasibility of different scenarios. They have developed their capacity for prospective analysis and the formulation of well-reasoned strategic recommendations. They understand the challenges of energy governance and are prepared to participate in an informed manner in energy debates as future citizens.

To deepen the reflection and open up perspectives for civic engagement, here are some key questions to explore collectively:

- **How do the lessons from our simulations apply to current energy debates in our country?** Applying the identified strategic principles to contemporary national energy issues. Critical analysis of current energy policies in light of the constraints discovered. Identifying points of convergence and divergence between simulations and political reality.
- **What democratic mechanisms allow citizens to influence energy choices?** Exploring instances of citizen participation in energy decisions (advisory councils, referendums, participatory budgets). Analyzing the role of elected officials, experts, and citizens in energy governance. Reflecting on the skills needed to participate in technical debates.

- **How to reconcile technical expertise and democratic legitimacy in energy choices?** A discussion of the tension between technical complexity and citizen accessibility of energy issues. An exploration of methods of scientific mediation and public debate. A reflection on the balance between delegation to experts and democratic control.
- **What personal contribution can you make to the energy transition beyond simulation?** Identifying accessible forms of civic engagement (associations, local projects, training options). Exploring the links between acquired skills and opportunities for action. Building a long-term vision of your role in tomorrow's energy society.

This final phase provides students with the analytical and strategic tools needed to become informed citizens, capable of evaluating energy policies and contributing constructively to democratic debates on the energy future of their society.

This modeling approach can be transferred to other systemic issues to develop in students a culture of quantitative analysis and foresight applied to societal challenges.



Going further



Smart microgrid project within the establishment

Students can design and implement a miniature energy system in their school, integrating renewable production (solar panels, small wind turbine), storage (batteries), and smart consumption management (programmable LED lighting, connected devices). This multidisciplinary project mobilizes the technical skills acquired during the construction of the simulator while confronting students with the practical challenges of real-time energy management. The educational objective is to transform the school into a living laboratory of the energy transition, where students concretely experiment with the principles of energy optimization and measure the impact of their choices on overall consumption.

Example: The Solar Schools program provides curriculum-aligned lessons and learning experiences for elementary and middle school students. Participating schools are equipped with energy meters that record consumption and production in real time. Teachers have access to a comprehensive toolbox and interactive digital tools to bring scientific concepts to life through real-world data analysis. <https://www.solarschools.net>



Territorial energy modeling with local authorities

Students extend their simulator to the scale of their municipality or inter-municipality in partnership with local technical services. They collect real-world energy data from the territory (consumption by sector, local energy mix, ongoing projects), adapt their model to territorial specificities, and propose contextualized energy transition scenarios. This collaboration with local public stakeholders allows students to understand the operational constraints of territorial energy planning and to measure the gap between theoretical modeling and practical implementation.

Example: The MITEC Launchpad is an initiative of the MIT Energy & Climate Club that offers MIT students the opportunity to solve real-world energy and climate problems with leading industry professionals. Students work on real-world energy modeling and climate planning projects in partnership with external organizations, developing practical solutions to contemporary energy challenges. <https://mitec-club.org>



Participation in international energy innovation competitions

Students deepen their expertise in energy modeling by participating in international competitions that reward technical innovation and scientific creativity. They develop functional prototypes of innovative energy systems (energy harvesting, consumption optimization, alternative storage) and present their solutions to a panel of experts. This participation enhances their technical skills while confronting them with the standards of contemporary energy innovation and the expectations of the professional world.

Example: The EUCYS (European Union Contest for Young Scientists) competition is open to young scientists aged 14 to 20. Participants can submit individual projects or projects in teams of up to three members, with a particular focus on energy and environmental innovation. Projects can be theoretical or experimental, and each country can submit up to three projects. This competition highlights originality and scientific curiosity, offering young talents the opportunity to share their research with an international audience.

https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/eucys_en



Running energy workshops in leisure centers

Students share their knowledge by leading fun energy workshops with younger children at local recreation centers. They adapt their simulator by creating serious games, organize simple scientific experiments, and raise awareness about energy efficiency through creative activities. This scientific outreach approach develops their teaching and communication skills while anchoring their learning through communication. It also helps spread energy culture throughout society.

Example: Student Energy (SE) is a global, youth-led organization empowering the next generation to accelerate the energy transition. Their Student Energy Chapters program allows students to create local chapters at their institutions. Participants develop concrete energy transition projects while benefiting from mentorship and an international peer network. This initiative fosters leadership development and creates a lasting impact that is passed on to new generations of students.

<https://studentenergy.org/chapters/>



Participation in citizens' assemblies on energy

Students engage in local participatory democracy initiatives dedicated to energy issues, bringing their technical expertise acquired during the construction of the simulator to inform citizen debates. They participate in local citizens' climate conventions, development councils, and advisory committees for territorial energy projects. This participation allows them to gain a concrete understanding of democratic energy decision-making processes and to measure the impact of citizen expertise on collective choices.

Example: The Youth Forum in Belgium is the official representative body for young people aged 16 to 30 in the Wallonia-Brussels Federation. Through participatory days, young people define concrete projects and issue official opinions on societal issues, particularly climate and energy issues. This structure allows young people to become agents of change by directly participating in decisions that affect them. <https://forumdesjeunes.be>



Creation of a citizen energy cooperative

Students are involved in the creation or development of a local energy cooperative, contributing their technical expertise to project evaluation, profitability analysis, and community outreach. They use their modeling skills to calculate the economic and environmental impacts of proposed facilities and participate in general meetings as members. This experience introduces them to collective energy entrepreneurship and alternative economic models for the energy transition.

Example: The SIREnergy project, funded by Interreg Europe, aims to promote citizen energy cooperatives as a model for local energy transition. During their event "Citizen Involvement through the Citizen Energy Cooperatives," they presented concrete examples of citizen participation in energy management and provided tools to involve young people in these initiatives. <https://www.interregeurope.eu/sirenergy>